

JIawei ZHANG

949-508-8986 • jerry.zhang0825@gmail.com • linkedin.com/in/shacozha • https://github.com/ShaCoZha

SUMMARY

First-year M.S. student in Computer Science at Northwestern University with a strong background in machine learning, deep learning, and interactive systems. Experienced in building end-to-end ML and DL pipelines, including data collection, feature engineering, transfer learning, and real-time inference, as well as developing interactive applications using Unity and AR/XR technologies. Interested in applying machine learning to human-computer interaction, with a focus on next-generation games and interactive systems.

EDUCATION

M.S. in Computer Science

Graduating June 2027

Northwestern University, IL

3.9 GPA

McCormick School of Engineering

Relevant coursework: Machine Learning and Sensing, Natural and Artificial Vision, Computer Graphics, Deep Reinforcement Learning

B.S. Software Engineering

2021 - 2025

University of California Irvine, Irvine, CA

3.829 GPA

Donald Bren School of Information and Computer Sciences, ICS Honors Program

Relevant coursework: Design and Analysis of Algorithms and Data Structure, Concepts in Programming Languages, Machine Learning

TECHNICAL SKILLS

Machine Learning: Model training, evaluation, and deployment through academic projects and research

Deep Learning Frameworks: Transfer learning and model fine-tuning using PyTorch

Graphics Knowledge and Programming: WebGL programming with custom shaders and JavaScript for interactive graphics

Unity Game Engine: 2D and 3D game development, gameplay systems, and interaction logic using Unity and C#

AR/XR Development: Development of interactive 3D AR/XR applications with spatial interaction

Programming in Python: Python programming for machine learning, data processing, and system integration

Programming in C++: C++ data structures, algorithms, and performance-oriented applications

Testing and Programming in Java: Backend development and testing in Java with JUnit 5

Programming in Next and React JS: Frontend development using Next.js and React for interactive web applications

Basic Science Backgrounds: Strong foundation in mathematics, physics, and statistics

ACADEMIC PROJECTS

ML-Driven Object Recognition and Spatial Interaction in Mixed Reality

Aug 2025 - Oct 2025

Developed a Quest Passthrough-based mixed reality object recognition system that streams camera frames to a Python server for ML inference and visualizes detected objects with world-aligned spatial annotations in real time.

- Built a real-time Quest Passthrough camera pipeline, streaming RGB frames to a Python inference server via WebSocket, receiving object detection results asynchronously, and managing low-latency bidirectional communication.
- Implemented spatially grounded object visualization in mixed reality, projecting 2D detection results onto the passthrough view, converting screen-space bounding boxes to world-space positions via raycasting, and rendering labeled, world-aligned annotations anchored in the physical environment.

Gaze and Head Gesture Control Application Development

Oct 2025 - Dec 2025

Developed a gaze and head gesture control application based on OpenGaze using PyTorch-based transfer learning for gaze tracking and head movement in a four-person team.

- Applied transfer learning to head movement tracking by fine-tuning pre-trained head feature model in PyTorch, improving robustness under user-specific variations.
- Built the application interface to integrate real-time gaze and head gesture inference with user interaction logic.

Real-Time Audio Event Recognition System Development

Oct 2025 - Dec 2025

Developed a real-time audio event recognition system that integrates traditional machine learning and deep learning models to perform live inference on streaming audio with feature extraction and model comparison visualization.

- Designed and implemented the full machine learning pipeline, including custom audio data collection, feature extraction (FFT, MFCC, RMS), model training, hyperparameter tuning, and cross-environment evaluation for acoustic activity recognition
- Trained and fine-tuned a deep learning model via transfer learning, adapting a VGGish/Ubioustics-based pre-trained model to a custom 5-class audio dataset using TensorFlow, and integrated the model into a real-time inference system for live audio recognition.

Revisiting and Re-implementing Class String

Jan 2024 - Jun 2024

Implemented and compared algorithms on the basis of re-implementation of C++ std class String with various data structures.

- Re-implemented the C++ std class String with data structures including linked lists, trees, heaps, and more.
- Implemented sorting and searching algorithms like binary search, BFS or DFS on such re-implementations

PROFESSIONAL EXPERIENCE

Edge Digger – Ludum Dare 57

Apr 2025 (3-day game jam)

game <https://ldjam.com/events/ludum-dare/57/cyber-digger-for-now>

Worked in a group of five to design and engineer a game with a given topic in three days.

- Programmed in Unity to implement the movement mechanism for the player, and the map obstacles generation mechanism for gameplay.
- Developed and implemented the survival and awarding mechanism and the corresponding UI for the game.

TapTap Spotlight Gamejam

Oct 2024 (30 days game jam)

repo <https://github.com/ShaCoZha/TapTap-Spotlight-ICUgame>

Collaborated with a group of game engineers and producers to participate in and complete a game project for the TapTap Spotlight Game Jam.

- Implemented player control, animation, and physics-based interaction systems using Unity and CSharp.
- Developed the interaction mechanism between the player and the environment, implementing mobile adaptations to players' operations.

Quadruple Root Technology, Jiangsu, China: Software Research Intern

Jun 2024 – Aug 2024

- Implemented the open-source RAG system QAnything with Qwen MML support. Made localized adjustments to the RAG retrieval algorithm. (Python Programming)
- Developed a front-end user interface for the localized RAG database Q/A system. Users can make quick Q/As with default database settings or formal Q/As by designating particular databases. (NextJS)

ClouDr. Group, Zhejiang, China: Software Engineering Intern

Sep 2024 - Oct 2024

- Participated in the Software Development process of an online medical platform, programmed and developed an customer-factory cost statistics back-end API and tested it with JUnit 5. (Java, JUnit 5)

ACADEMIC RESEARCH

Dual-IMU 6DoF World Lock Application in Augmented Reality (Northwestern University)

Oct 2025 - Ongoing

Advised by **Professor Karan Ahuja**

- Implemented a Python-to-AR communication interface, enabling the AR application to receive real-time 6DoF prediction results from a Python server and drive world-locked spatial anchors using dual-IMU input.
- Designed and developed interactive AR application logic, including calibration workflow, spatial anchor interactions, gesture-based menu activation, object spawning (music, calendar, photo, timer), movement confirmation, and visual distance feedback for held objects.